Lab 2 - Knowledge Is Power (KIP) Product Specification Outline

Rachel Thomas

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Instructor: Thomas J. Kennedy

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1. Introduction

This section introduces the foundational purpose, scope, and structural overview of this technical specification.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the complete, formal functional and non-functional requirements for the Knowledge Is Power (KIP) prototype. This document defines an objective, baseline technical blueprint for developers, quality assurance testers, and project stakeholders to ensure verification, compliance, and successful implementation of all system architectural components.

1.2 Scope

KIP is a web-based educational platform engineered to provide young adults with an accessible framework for legal literacy regarding workplace, housing, and digital environments. Rather than requiring users to interpret abstract statutory text, KIP implements scenario-based decision trees and interactive exercises to deliver localized supplemental education. The primary objective of the core prototype is to demonstrate operational feasibility, evaluate user interactions, and analyze knowledge retention metrics.

1.3 Definitions, Acronyms, and Abbreviations

- API: Intermediary that takes requests from the Angular web screens, fetches the correct legal learning data from the server, and delivers it back to the user.
- CDC: Centers for Disease Control and Prevention.
- HTML5 / Bootstrap / JavaScript: The presentation layer tech stack used for building the mobile-friendly web interface.
- Java / JDBC: The application layer tech stack used to power the scenario, authentication, and curation engines.
- JDL (JHipster Domain Language): Formatting language used to outline an application's database structure in a single file.
- JWT (JSON Web Token): A secure, encrypted digital badge used to track user logins.
- KIP: Knowledge Is Power digital legal literacy platform.
- MFCD: Major Functional Component Diagram detailing system architecture layout.
- PostgreSQL / MySQL: The relational database management systems utilized by KIP. The development environment leverages MySQL for local database management and testing, while the production layer utilizes PostgreSQL to ensure robust, enterprise-grade data persistence.
- REST (Representational State Transfer): An architectural style for designing networked applications that relies on a stateless, client-server communication protocol, typically HTTP.
- VPS (Virtual Private Server): A private, isolated section of a physical server acting as an independent computer.

1.4 References

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1.5 Overview

The remainder of this document outlines the operational framework of the KIP application. Section 2 provides a high-level architectural overview, target user profiles, system constraints, and development assumptions. Section 3 defines specific functional requirements and technical specifications for individual system features.

2. Overall Description

This section outlines the high-level architectural framework, core features, and target user classifications for the platform.

2.1 Product Perspective

KIP operates as a standalone web application requiring only a standard web browser for user access. The system architecture isolates user interaction, business logic, and data storage into three distinct, modular layers:

- The Frontend (Presentation): The visual screens built with React or Angular, Bootstrap, and JHipster. It is designed to feel distinctly non-instutional, while remaining navigational for a young, technically aware demographic.
- The Backend (Application): The server-side logic layer that processes game rules, runs the branching story paths, and handles session management with execution efficiency in mind.
- The Database (Data): The storage center that keeps track of user accounts, scores, and game questions.

2.2 Product Functions

The main objectives of the KIP prototype are:

- Authentication Management: Verifies registered user credentials alongside a functional "Guest Mode" configuration utilizing localized session activity.
- Module Navigation: Enables users to select specific learning paths across multiple legal layers, including traffic regulations and digital copyright compliance.

- "Fact or Fiction?" Evaluation Engine: An interactive quiz framework processing true, false, and context-dependent parameters to assess user assumptions.
- "Branching Narrative" Player: A scenario application that dynamically parses multi-option decision forks to visualize the real-world legal consequences of user choices.
- Performance Tracking and Analytics: A background engine that measures performance to algorithmically suggest specific modules targeting the user's deficiencies.

2.3 User Characteristics

There are three types of people who will use KIP:

- Guest Users: Unregistered, casual participants utilizing localized sessions with limited data persistence to test platform mechanics.
- Registered Students: Primary end users who authenticate securely to save progress metrics, track previous performance changes, and access personalized dashboard recommendations.
- System Administrators: Designated technical team members responsible for testing, validation, and operational monitoring during the prototype deployment phase.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A